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Maths puzzles for early finishers

# Early Finisher Puzzles

## Primary 1

## Semester 1

Terms 1 and 2

20 single-page puzzles
Almost every puzzle checks itself
One star and two star pages
Teacher answer key included

This booklet belongs to

_____

[Publisher line: wording to be confirmed by the owner]

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# Teacher notes (DRAFT)

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## What this is

Twenty single-page maths puzzles for Primary 1 pupils who finish their work early. This is Semester 1 of two booklets (Terms 1 and 2). Each page stands alone, so you can hand out one page or keep a tray of pages. The pupil needs no help to start.

## Which terms

The pages are grouped by the order in which Primary 1 topics are typically taught: this booklet follows Terms 1 and 2, the other booklet follows Semester 2 (Terms 3 and 4). Schools vary, so check your scheme of work: use a page whenever its topic has been taught, from either booklet.

## Stars

One star pages practise a topic as it is usually taught (10 pages). Two star pages use the same topics with two steps or two clues to combine (10 pages).

## How the pages check themselves

The boxed strip at the foot of each page tells the pupil how to check. The check shows that something is wrong. It does not give the answer, and it asks the pupil to see whether the answer makes sense. On the Finish the Figure page the check tests whether the pupil can tell which printed half matches; look at the pupil's drawing at a glance.

- **Code word.** The answers give letters that spell a word. The strip gives no hint of the word. Each word can be spelt from its letters in one way only, so answers in the wrong boxes never make another word.
- **Colour by code.** Each answer is found in a key of five numbers and the part is coloured with that crayon. The pupil counts the parts of each colour and compares with the box on the page; matching parts must share a colour.
- **Route.** The line must reach FINISH through the stated number of boxes, and every box on it must follow the rule on the page.
- **Logic grid, queue, strips and matching pictures.** Each number, child, strip or picture has a letter; the letters, read in the order asked, must spell a word. Where one item is left out, its letter is a decoy that is not part of the word.
- **Total.** The whole count must make the total printed in the strip (the graph pages add a second sum).
- **Loop.** The numbers on touching tiles must match all the way round the loop.

Marking is not needed. To spot-check, look only at the check strip.

## How the checks were tested

A script tried every single wrong answer and every wrong arrangement on every page. Each check rejected at least 95% of them. A wrong answer that spells any of about 129 000 English words counts as not rejected. Two wrong answers at once were not all tested.

# Teacher notes: using the pages (DRAFT)

## Using the pages

- **Time.** Most pages take 5 to 15 minutes; the colour-by-code pages (S1-02, S1-04) take longer because pupils colour a picture. Pupils need their own crayons (five colours: yellow, red, blue, green, orange). For a pencil-only copy the pupil writes Y, R, B, G or O in each part; the tally and the matching parts are checked the same way. Light crayons work best: a dark crayon hides the number in the part on a photocopy.
- **If the check fails.** The pupil goes back, looks for the slip, and asks a friend or you if still stuck.
- **Problem solving.** Some pages (S1-12) practise the kinds of question that appear in school tests, such as comparing amounts and making them equal, rather than one numbered syllabus outcome. The key tags them problem solving.
- **Stars.** One star: one skill, as usually taught. Two stars: the same topics with two steps or two clues to combine.

## Photocopying

All pages are black and white with solid lines, so they copy cleanly. Each page is one side only. Pupils write on the page: the boxes are large for young hands. Use A4 and print at 100% (actual size); on the ruler pages a real ruler matches the drawn ruler only at 100%.

# Contents: Semester 1


| Page | Code | Puzzle | Tier |
|---|---|---|---|
| S1-01 | P1-25 | Count the Crowd | 1 star |
| S1-02 | P1-01 | Colour the Rocket | 1 star |
| S1-03 | P1-24 | Words and Numbers | 1 star |
| S1-04 | P1-12 | Colour the House | 1 star |
| S1-05 | P1-19 | Shape Count | 1 star |
| S1-06 | P1-07 | The Queue | 2 stars |
| S1-07 | P1-08 | Make Twelve | 1 star |
| S1-08 | P1-37 | Bonds and Families | 2 stars |
| S1-09 | P1-10 | Take Away Code | 1 star |
| S1-10 | P1-21 | Story Sums | 2 stars |
| S1-11 | P1-09 | Magic Square | 2 stars |
| S1-12 | P1-38 | More or Fewer | 2 stars |
| S1-13 | P1-02 | Mystery Numbers | 2 stars |
| S1-14 | P1-04 | Sticker Grid | 2 stars |
| S1-15 | P1-26 | Chart Patterns | 2 stars |
| S1-16 | P1-33 | Shape Builder | 2 stars |
| S1-17 | P1-34 | Finish the Figure | 2 stars |
| S1-18 | P1-31 | Shortest to Longest | 1 star |
| S1-19 | P1-17 | How Long? | 1 star |
| S1-20 | P1-13 | Equal Groups | 1 star |


Pages are independent. The footer code shows the page in this booklet and, in brackets, its number in the full set of 40 (P1-01 to P1-40).

## What the booklet covers

**Whole Numbers 1.1** counting to tell the number of objects in a given set **S1-01**

**Whole Numbers 1.2** number notation, representations and place values (tens, ones) **S1-02, S1-13, S1-15**

**Whole Numbers 1.3** reading and writing numbers in numerals and in words **S1-03**

**Whole Numbers 1.4** comparing the number of objects in two or more sets **S1-01**

**Whole Numbers 1.5** comparing and ordering numbers **S1-12, S1-13, S1-14, S1-15**

**Whole Numbers 1.6** patterns in number sequences **S1-15**

**Whole Numbers 1.7** ordinal numbers (first, second, up to tenth) and symbols (1st, 2nd, 3rd, etc.) **S1-03, S1-06**

**Addition and Subtraction 2.1** concepts of addition and subtraction **S1-07, S1-08, S1-09, S1-10, S1-12**

**Addition and Subtraction 2.2** use of +, - and = **S1-07, S1-08, S1-10**

**Addition and Subtraction 2.3** relationship between addition and subtraction **S1-08, S1-09**

**Addition and Subtraction 2.4** adding more than two 1-digit numbers **S1-11**

**Addition and Subtraction 2.7** mental calculation involving addition and subtraction (within 20; 2-digit number and ones without renaming; 2-digit number and tens) **S1-07, S1-09, S1-10**

**Multiplication and Division 3.1** concepts of multiplication and division **S1-20**

**Multiplication and Division 3.2** use of x **S1-20**

**Multiplication and Division 3.3** multiplying within 40 **S1-20**

**Length 1.1** measuring length in centimetres **S1-19**

**Length 1.2** use of abbreviation cm **S1-18, S1-19**

**Length 1.3** comparing and ordering lengths in cm **S1-18**

**2D Shapes 1.1** identifying, naming, describing and classifying 2D shapes (rectangle, square, triangle, circle, half circle, quarter circle) **S1-04, S1-05, S1-16**

**2D Shapes 1.2** forming different 2D figures with rectangle, square, triangle, half circle, quarter circle **S1-16**

**2D Shapes 1.3** identifying the 2D shapes that make up a given figure **S1-05, S1-16**

**2D Shapes 1.4** copying figures on dot grid or square grid **S1-17**

Outcomes practised in the other booklet: Addition and Subtraction 2.5; Addition and Subtraction 2.6; Multiplication and Division 3.4; Money 1.1; Length 1.4; Time 2.1; Time 2.2; Time 2.3; Time 2.4; Picture Graphs 1.1.

S1-01 (P1-25)

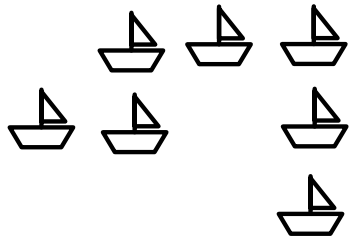
# Count the Crowd

★ Name _____

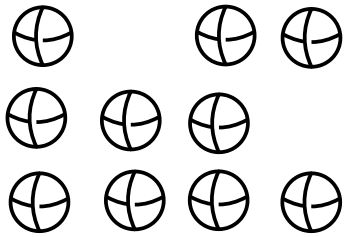
**Topic: Counting a set and comparing two sets**

Count each set.
Match the pictures with lines.
Use the code table.

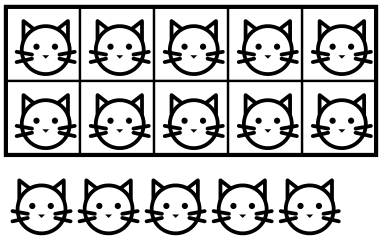
1 How many boats?



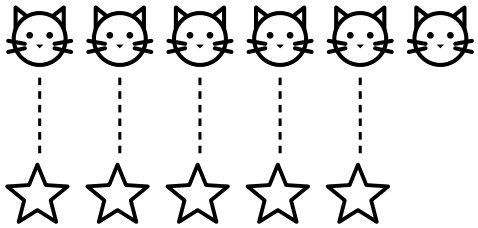
2 How many balls?



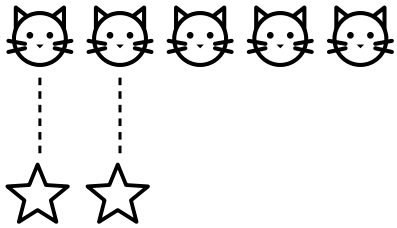
3 How many cats?



4 How many more cats than stars?



5 How many fewer stars than cats?



**Code table**

| | | | | | |
|---|---|---|---|---|---|
| 1 = T | 3 = H | 5 = Y | 6 = F | 7 = C | 8 = Q |
| 10 = L | 15 = O | | | | |

**Check it.** Write the letters in order. Do they spell a word? ☐☐☐☐☐

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S1-02 (P1-01)

# Colour the Rocket

Name _____

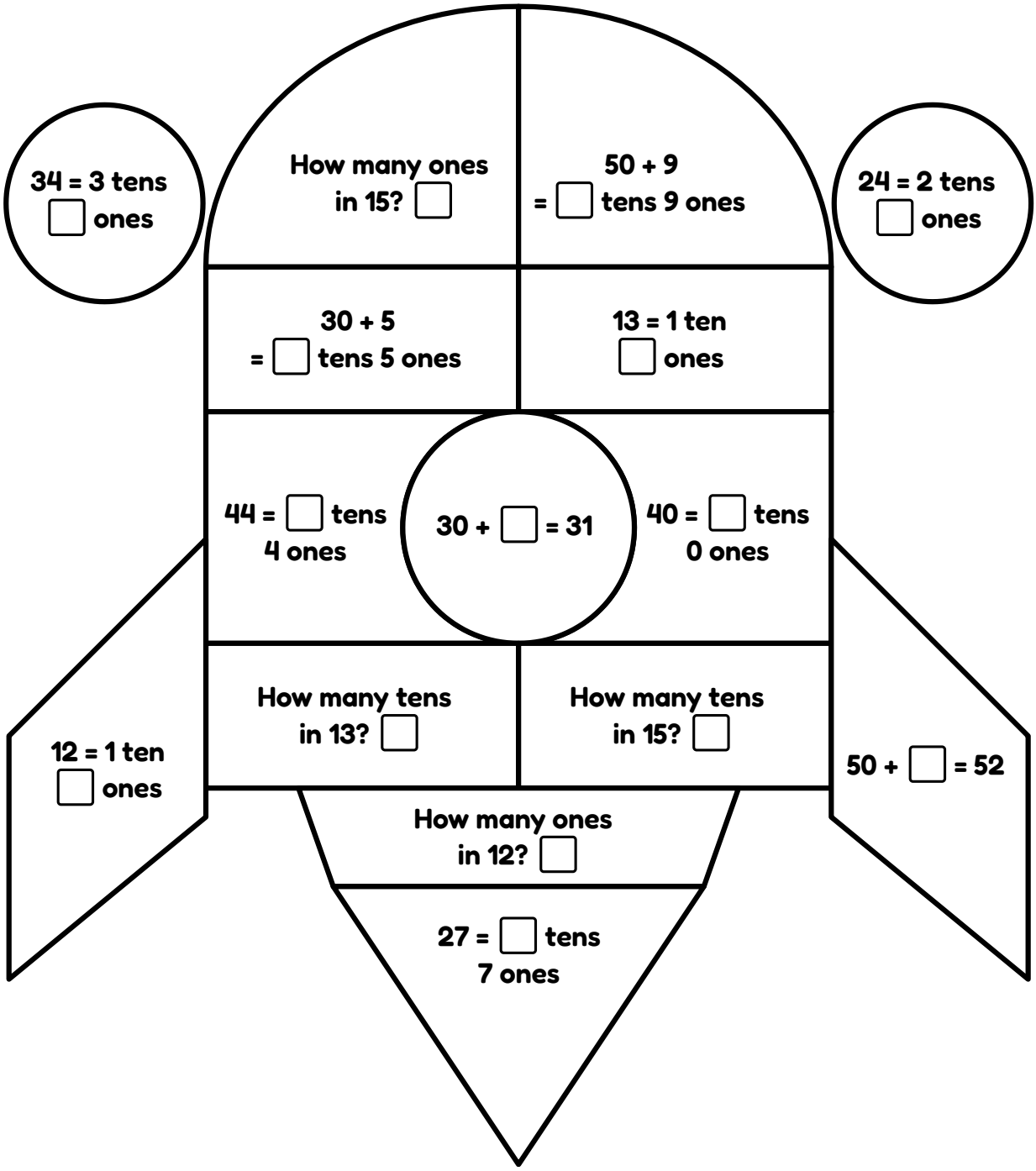
**Topic: Place value: tens and ones**

Find the missing number in each part.

Find it in the key.

Colour the part. No crayon? Write Y, R, B, G or O.

| 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|
| yellow | red | blue | green | orange |




**Check it.**

Count the parts of each colour. You should have: yellow 3, red 4, blue 2, green 4, orange 2.

Parts in the same place on the left and on the right have the same colour.

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S1-03 (P1-24)

# Words and Numbers

★ Name _____

**Topic: Number words, numerals and ordinals**

Start at START.
Go to the box saying your number.
Move up, down, left or right.

| | | | |
|---|---|---|---|
| START<br>**ninety-five**<br>**17** | **seventeen**<br>**9th** | **fourth**<br>**29** | **twenty-nine**<br>**94** |
| **thirty**<br>**4th** | **ninth**<br>**31** | **thirty-one**<br>**4th** | **ninety-four**<br>**38** |
| **forty-seven**<br>**39** | **thirty-one**<br>**1st** | **fourth**<br>**3rd** | FINISH<br>**thirty-eight**<br>**5th** |
| **forty-nine**<br>**94** | **tenth**<br>**41** | **eighth**<br>**3rd** | **forty-one**<br>**57** |

**Check it.** Does your line reach FINISH? Count its boxes: are there 8? Does each top word match?

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S1-04 (P1-12)

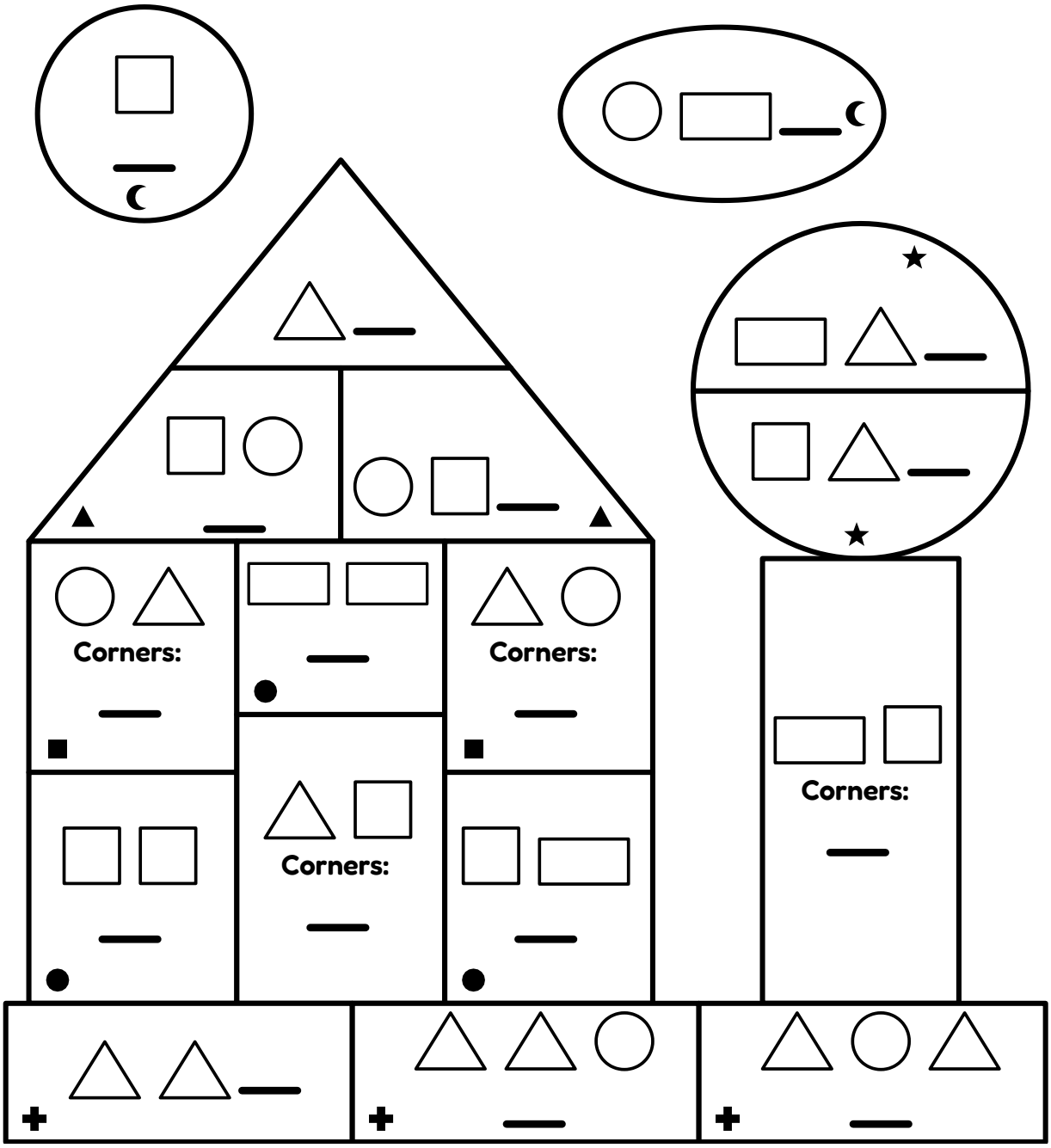
# Colour the House

Name _____

**Topic: Shapes: corners**

Count the corners. Write it on the thick line.
Find the number in the key.
Colour the part. No crayon? Write Y, R, B, G or O.

| 3 | 4 | 6 | 7 | 8 |
|---|---|---|---|---|
| yellow | red | blue | green | orange |




**Check it.**
Count the parts of each colour. You should have: yellow 3, red 4, blue 3, green 3, orange 4.

Parts with the same small mark have the same colour:

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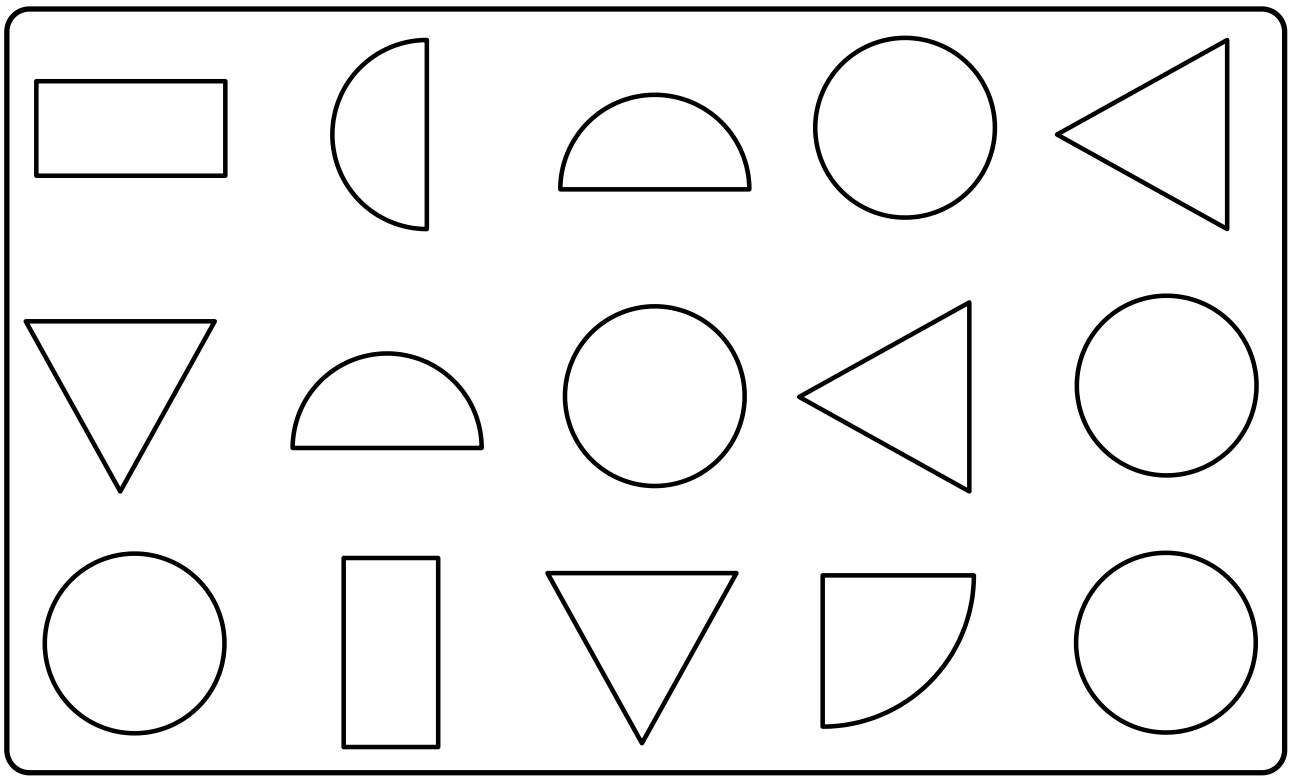
S1-05 (P1-19)

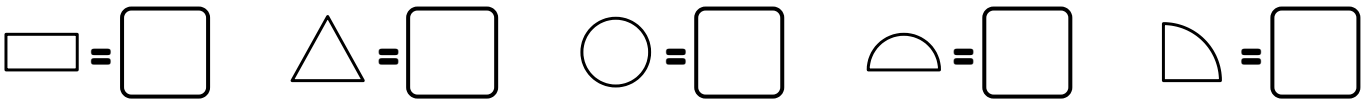
# Shape Count

Name _____

**Topic: 2D shapes: rectangle, triangle, circle, half circle, quarter circle**

Count each kind of shape in the picture.
Write how many in the box.
Use the code table.





**Code table**

| | | | | | |
|---|---|---|---|---|---|
| 1 = D | 2 = B | 3 = L | 4 = U | 5 = I | 6 = Q |
| 7 = H | 8 = S | | | | |

**Check it.** Do your five numbers add to 15? Write the letters. Do they spell a word?

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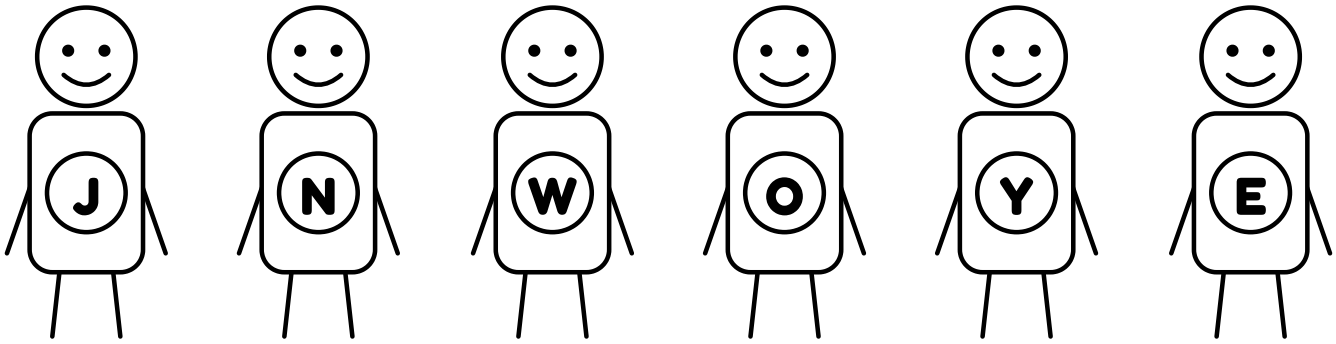
S1-06 (P1-07)

# The Queue

★★ Name _____

**Topic: Ordinal numbers: 1st to 5th, next to**

Six children. Five join the queue.
Read the clues.
Write each name in its place.




**The queue has 5 places. The front is 1st.**

**1st** ☐

**2nd** ☐

**3rd** ☐

**4th** ☐

**5th** ☐

- Kumar is not in the queue.
- Ali is 3rd in the queue.
- Ali is next to Aisha.
- Lily is 5th in the queue.
- Lily is next to Ravi.

**Check it.** Read the shirt letters from 1st to 5th. Do they spell a word? ☐☐☐☐☐

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S1-07 (P1-08)

# Make Twelve

Name _____

**Topic: Addition within 20**

Start at START.
Step only on sums that make 12.
Move up, down, left or right.

| | | | |
|---|---|---|---|
| 6 + 6 | 6 + 6 | 5 + 7 | 6 + 4 |
| 9 + 3 | 2 + 6 | 9 + 3 | FINISH<br>4 + 8 |
| 5 + 7 | 6 + 2 | 8 + 4 | 9 + 5 |
| START<br>8 + 4 | 3 + 9 | 9 + 5 | 4 + 7 |

**Check it.** Does your line reach FINISH? Count its boxes: are there 8? Does every box make 12?

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S1-08 (P1-37)

# Bonds and Families

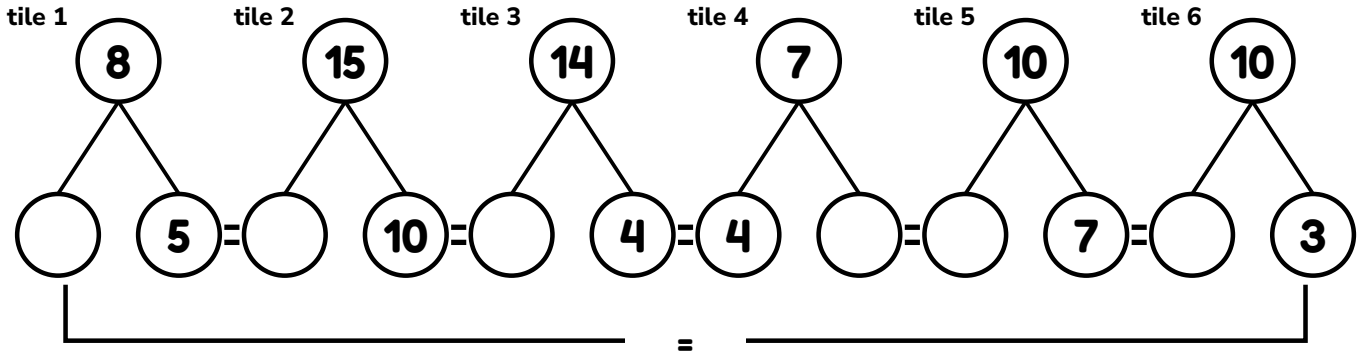
★★ Name _____

**Topic: Number bonds to 20: parts, wholes and the loop that closes**

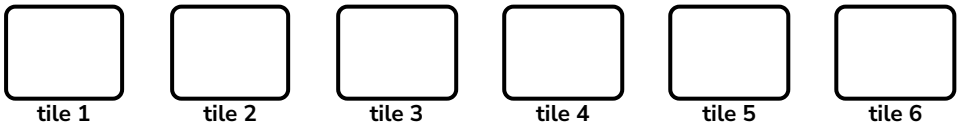
Find the number in each empty circle.
Touching circles joined by = match.
Go round the loop.

**Example**
Whole 9. Left part 4. Right part?
9 − 4 = 5. Check: 4 + 5 = 9.

**Each tile: the top number is the whole. The two below are its parts.**




**Write the numbers for the empty circles, tile 1 to tile 6.**




**Check it.**
Go round the loop. Do the numbers joined by = match?

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S1-09 (P1-10)

# Take Away Code

Name _____

**Topic: Subtraction within 20; missing numbers**

Find the missing numbers.
Use the code table.

① 18 – 9 = ☐

② 15 – 8 = ☐

③ 8 – 4 = ☐

④ 10 – 2 = ☐

⑤ 17 – ☐ = 12

⑥ 19 – ☐ = 3

**Code table**

| | | | | | |
|---|---|---|---|---|---|
| 2 = J | 3 = V | 4 = U | 5 = I | 7 = O | 8 = S |
| 9 = C | 15 = L | 16 = N | 18 = M | 19 = Q | |

**Check it.** Write the letters in order. Do they spell a word?

☐☐☐☐☐☐

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S1-10 (P1-21)

# Story Sums

★★ Name _____

**Topic: Add and take away stories within 20**

Read each story.
Write how many.
Use the code table.

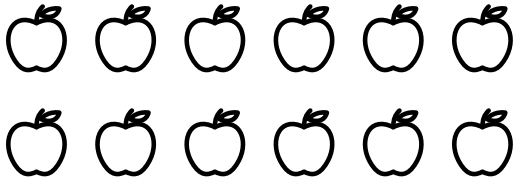
**1** Ali has 3 cats.
Ali has 3 fewer than Ravi.
How many does Ravi have?



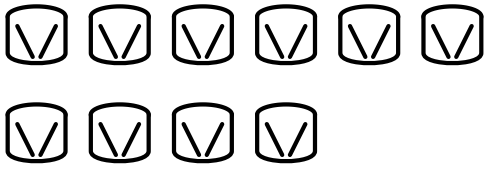
**2** Ravi has 3 flowers.
Ali has 2 more than Ravi.
How many does Ali have?



**3** Mei has 6 apples.
Lily has 6.
How many in all?



**4** Ben has 6 drums.
Siti has 4.
How many more does Ben have?



**5** Aisha has 6 balls.
Aisha loses 3.
How many are left?



**6** Lily has 5 stars.
Lily finds 9 more.
How many now?



**Code table**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0 = B | 1 = R | 2 = K | 3 = E | 5 = O | 6 = M | 12 = N | 14 = Y |

**Check it.** Write the letters in order.
Do they spell a word?

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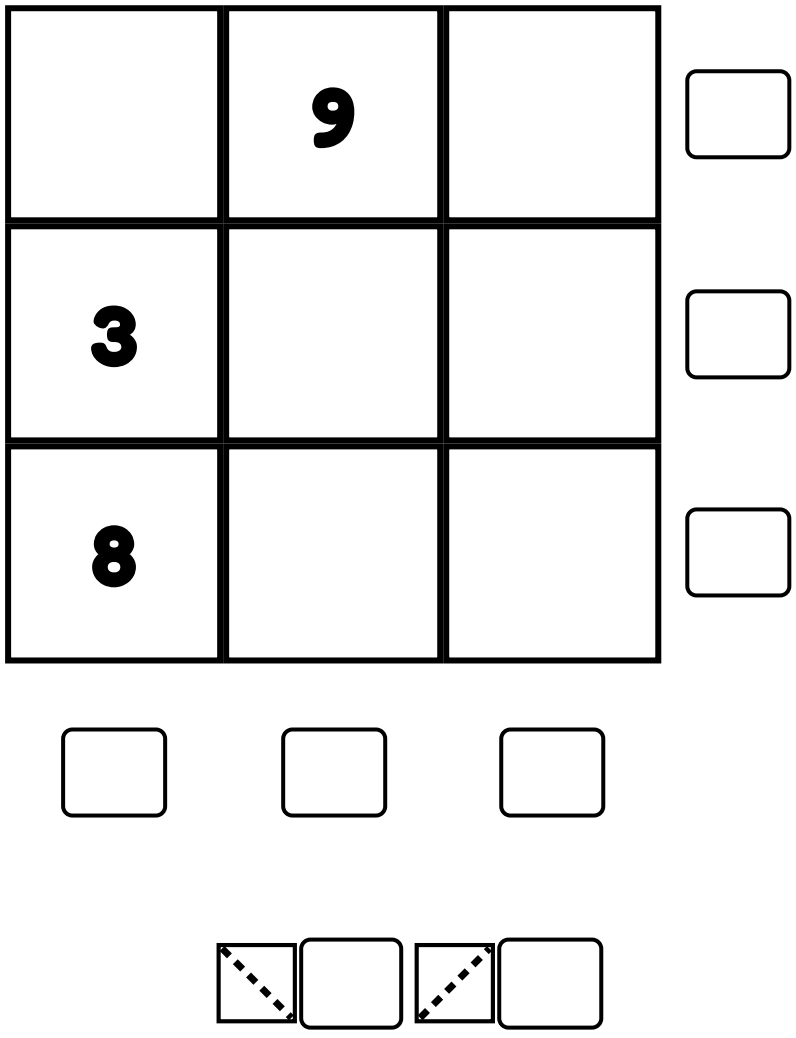
S1-11 (P1-09)

# Magic Square

★★ Name _____

**Topic: Adding three 1-digit numbers**

Use the numbers 1 to 9 once.
Every line across and down makes 15.
Both lines from corner to corner make 15.




**Numbers to use. Cross out each one you use.**

1 2 3 4 5 6 7 8 9

**Check it.** Write each line total in the small boxes. Are all eight totals 15?

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S1-12 (P1-38)

# More or Fewer

★★ Name _____

**Topic: Comparing amounts: more than, fewer than and making the same**

Read the clues.
Use the grid to cross and tick.
Write each amount, then its letter.

**Six amounts of stickers. Five children.**

| 5 R | 7 O | 8 L | 16 M | 19 Y | 20 G |
|---|---|---|---|---|---|

- Nobody has 16.
- Siti has 2 more than Ben.
- Ben has 3 fewer than Mei.
- Kumar gives 7 stickers to Ben. Then they have the same number.

**How to use the grid**

| | 5 | 9 |
|---|---|---|
| **Zed** | ✓ | ✗ |
| **Yan** | ✗ | |

Clue: Zed has 5.
Tick Zed under 5.
Cross out the rest of that line across and down.

| | 5 | 7 | 8 | 16 | 19 | 20 |
|---|---|---|---|---|---|---|
| **Ali** | | | | | | |
| **Mei** | | | | | | |
| **Siti** | | | | | | |
| **Ben** | | | | | | |
| **Kumar** | | | | | | |

| | number | letter |
|---|---|---|
| **Ali** | | |
| **Mei** | | |
| **Siti** | | |
| **Ben** | | |
| **Kumar** | | |

**Check it.** Read the five letters down your list. Do they spell a word?

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S1-13 (P1-02)

# Mystery Numbers

★★ Name _____

**Topic: Numbers to 100: tens digit, ones digit, more and less**

Find each mystery number.
Write it in the box.
Use the code table.

1. My ones digit is 6.
   I am less than 73.
   I am more than 59. ☐

2. My ones digit is 5.
   I am less than 95.
   I am more than 82. ☐

3. My tens digit is more than 2.
   I am less than 38.
   My ones digit is 6. ☐

4. I am less than 61.
   My ones digit is 6.
   My tens digit is more than 4. ☐

5. I am more than 23.
   My ones digit is 6.
   I am less than 34. ☐

**Code table**

| | | | | | |
|---|---|---|---|---|---|
| 26 = E | 34 = B | 36 = I | 46 = D | 56 = C | 65 = N |
| 66 = J | 67 = M | 85 = U | 95 = X | | |

**Check it.** Write the letters in order. Do they spell a word? ☐☐☐☐☐

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S1-14 (P1-04)

# Sticker Grid

★★ Name _____

**Topic: Comparing numbers to 100 (more, fewer, greatest, smallest)**

Read the clues.
Use the grid to cross and tick.
Write each number, then its letter.

**Six numbers of stickers. Five children.**

| 12 R | 23 P | 63 I | 78 T | 88 U | 99 F |
|---|---|---|---|---|---|

- Nobody has the number 23.
- Kumar has more than Ben.
- Ali has the greatest number.
- Siti has the second greatest number.
- Mei has fewer than Ben.

**How to use the grid**

| | 5 | 9 |
|---|---|---|
| **Zed** | ✓ | ✗ |
| **Yan** | ✗ | |

Clue: Zed has 5.
Tick Zed under 5.
Cross out the rest of that line across and down.

| | 12 | 23 | 63 | 78 | 88 | 99 |
|---|---|---|---|---|---|---|
| **Ali** | | | | | | |
| **Mei** | | | | | | |
| **Siti** | | | | | | |
| **Ben** | | | | | | |
| **Kumar** | | | | | | |

| | number | letter |
|---|---|---|
| **Ali** | | |
| **Mei** | | |
| **Siti** | | |
| **Ben** | | |
| **Kumar** | | |

**Check it.** Read the five letters down your list. Do they spell a word?

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S1-15 (P1-26)

# Chart Patterns

★★ Name _____

**Topic: Number patterns on a chart: 10 more, 10 less, and a growing table**

Use the chart and the table.
Write the missing numbers.
Use the code table.

**1 and 2: fill the empty boxes, reading order.**

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| 41 | 42 | | 44 | 45 | 46 | 47 | 48 | 49 | 50 |
| 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 |
| 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 |
| 71 | 72 | 73 | 74 | 75 | | 77 | 78 | 79 | 80 |

(3) 10 more than 60 is ☐ (4) 10 less than 54 is ☐

**5 and 6: Mei gets sweets each day. Write the missing numbers.**

| Day 1 | Day 2 | Day 3 | Day 4 | Day 5 | Day 6 |
|---|---|---|---|---|---|
| 3 | 6 | ☐ | 12 | ☐ | 18 |

**Code table**

| | | | | | |
|---|---|---|---|---|---|
| 9 = U | 15 = T | 17 = Q | 25 = J | 43 = P | 44 = N |
| 53 = R | 54 = X | 70 = A | 71 = Z | 76 = E | |

**Check it.** Write the letters in order. Do they spell a word? ☐☐☐☐☐☐

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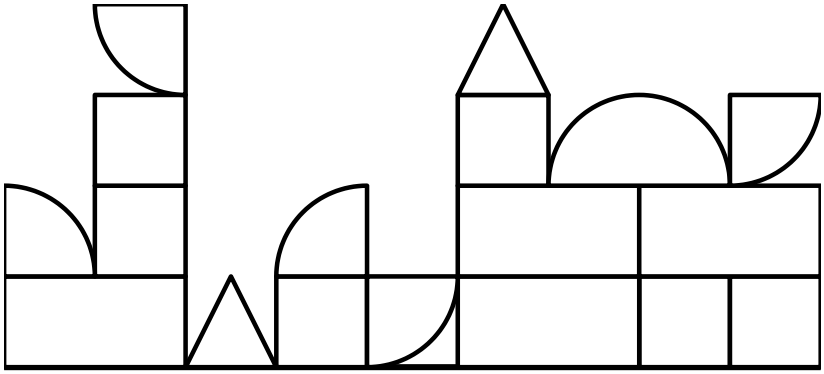
S1-16 (P1-33)

# Shape Builder

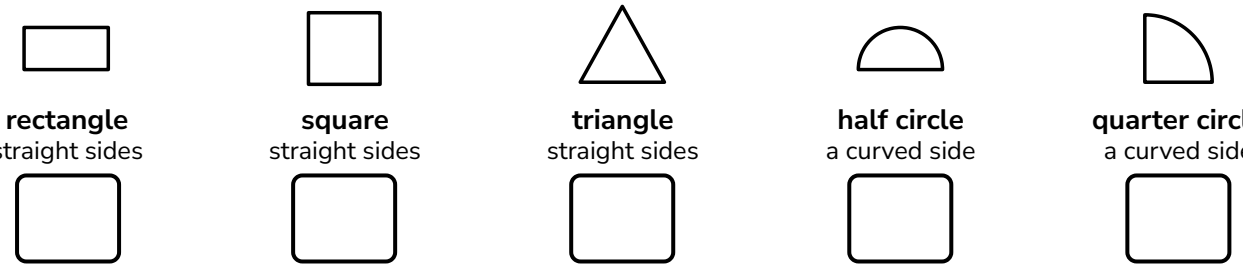
★★ Name _____

**Topic: 2D shapes: count the shapes in a figure and form a figure**

Count each kind of shape.
Which pieces form the small figure?
Use the code table.



**Squares and rectangles are different kinds here.**



**rectangle**
straight sides

**square**
straight sides

**triangle**
straight sides

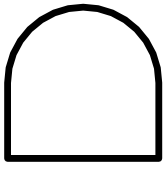
**half circle**
a curved side

**quarter circle**
a curved side

**Code table**

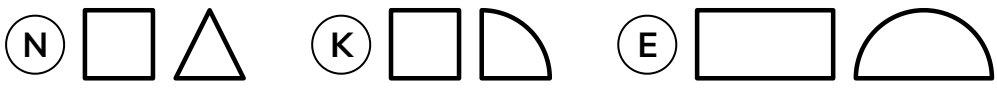
| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 = G | 2 = D | 3 = X | 4 = B | 5 = I | 6 = U | 7 = K | 8 = R |

**make this figure**



Which pieces make this figure?

Write that letter last.




**Check it.** Do the five numbers add to 18? Write six letters. Do they spell a word?

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S1-17 (P1-34)

# Finish the Figure

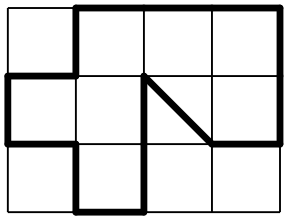
★★ Name _____

**Topic: Copy figures on a square grid**

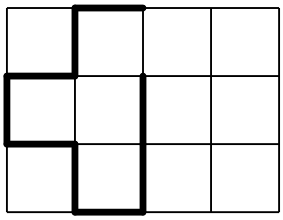
Copy the figure on the grid.
Finish the right half.
Find the matching right half.

**1**

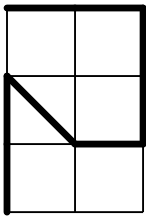
copy this



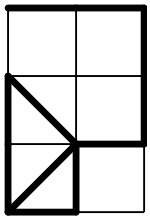
finish it here



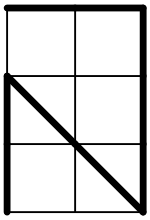
which right half matches?



F



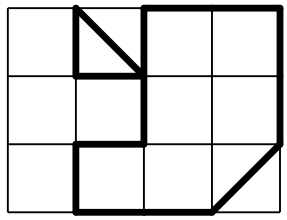
M



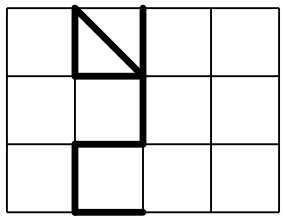
H

**2**

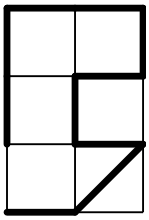
copy this



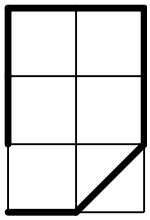
finish it here



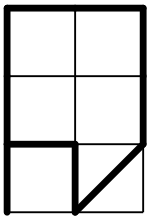
which right half matches?



V



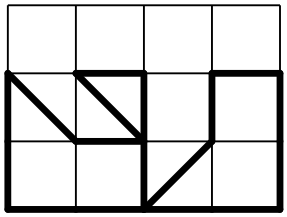
R



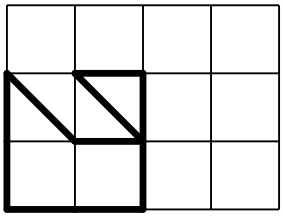
K

**3**

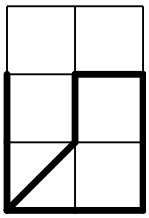
copy this



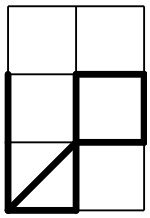
finish it here



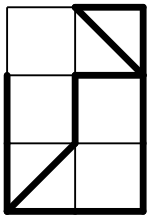
which right half matches?



O



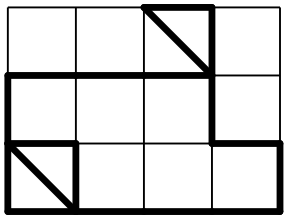
B



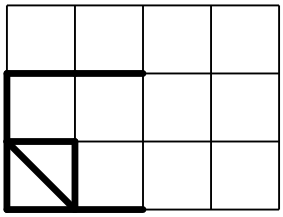
S

**4**

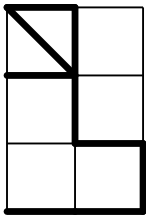
copy this



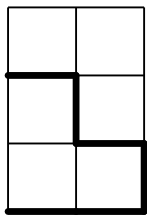
finish it here



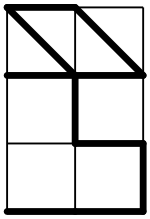
which right half matches?



G



X



Z



**Check it.** Write each matching letter. Do they spell a word?

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S1-18 (P1-31)

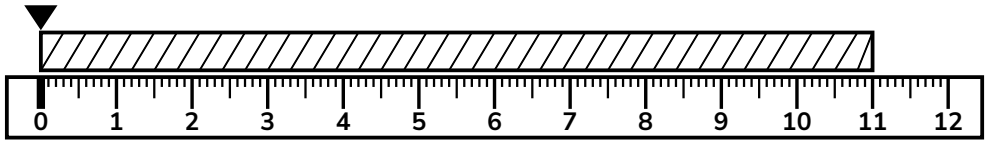
# Shortest to Longest

★ Name _____

**Topic: Length in cm: measure and put in order**

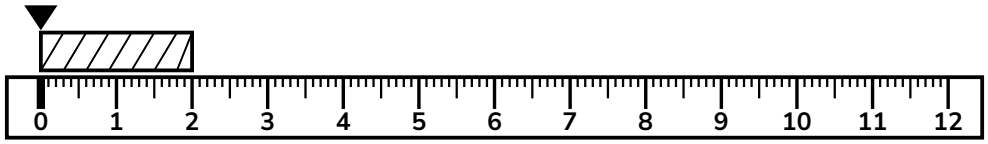
Measure each strip in cm.
Order the strips, shortest first.
Write the letters. Leave out the longest.

**T**



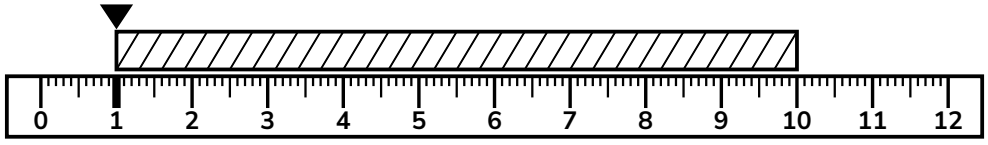

☐ cm

**B**



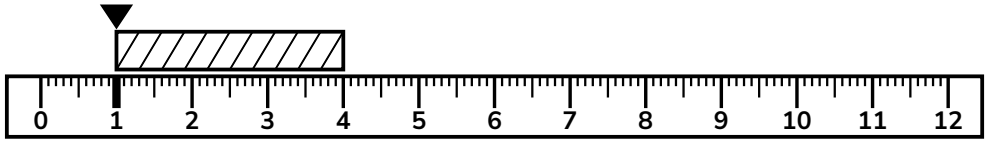

☐ cm

**N**



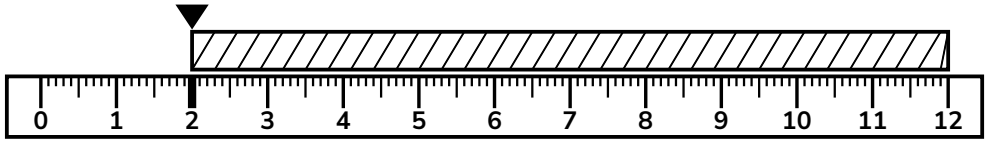

☐ cm

**L**



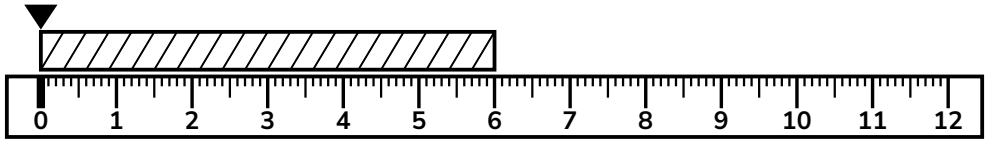

☐ cm

**K**




☐ cm

**A**




☐ cm

**Check it.** Read your five letters, shortest strip first. Do they spell a word? ☐ ☐ ☐ ☐ ☐

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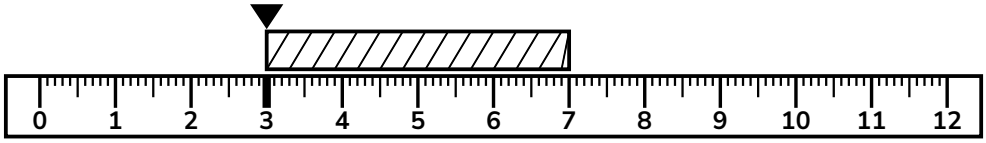
S1-19 (P1-17)

# How Long?

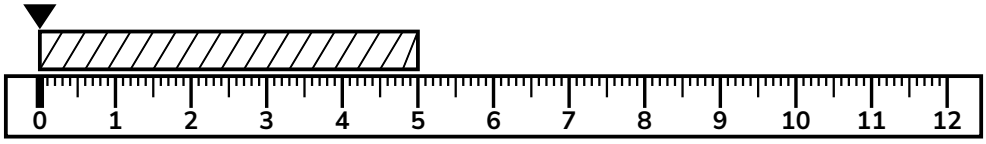
★ Name _____

**Topic: Length: measure in centimetres**

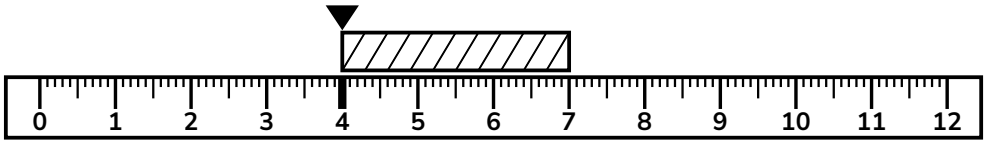
Each strip starts at the arrow.
Count the cm to the other end.
Write the length. Use the code table.

1 

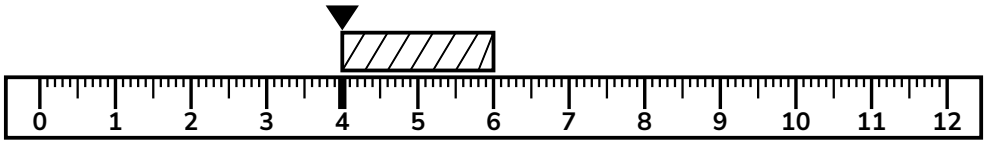
☐ cm

2 

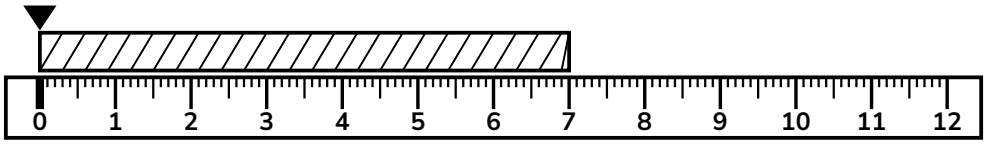
☐ cm

3 

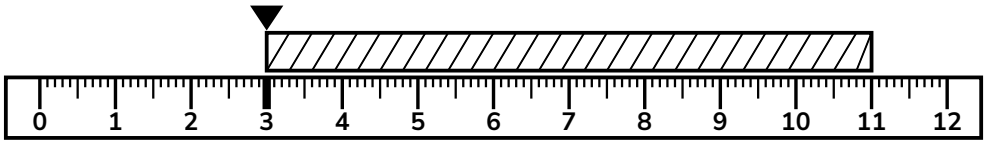
☐ cm

4 

☐ cm

5 

☐ cm

6 

☐ cm

**Code table**

| | | | | | |
|---|---|---|---|---|---|
| 1 = U | 2 = G | 3 = N | 4 = L | 5 = E | 6 = X |
| 7 = T | 8 = H | 9 = J | 10 = Q | | |

**Check it.** Write the letters in order. Do they spell a word? ☐☐☐☐☐☐

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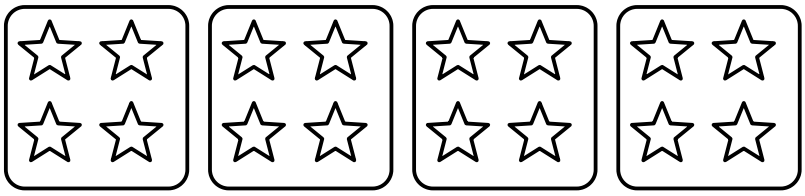
S1-20 (P1-13)

# Equal Groups

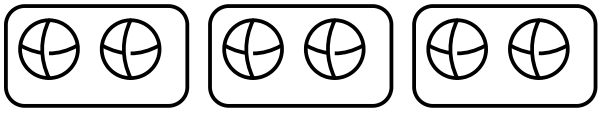
★ Name _____

**Topic: Multiplication: equal groups and repeated addition**

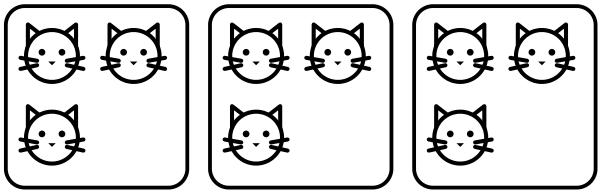
Each bag has the same number of things.
Write the bags, the things and the total.
Use the code table.

1 

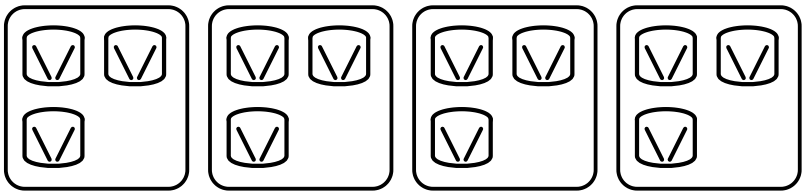
☐ bags × ☐ things = ☐ total

2 

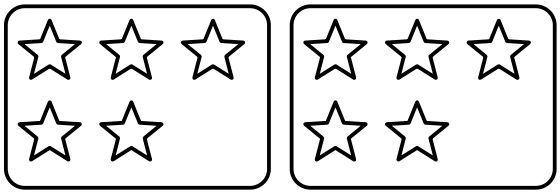
☐ bags × ☐ things = ☐ total

3 

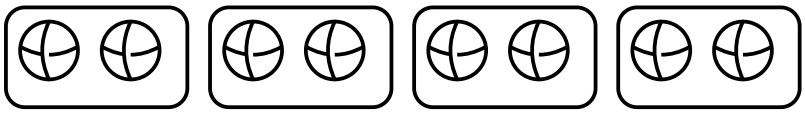
☐ bags × ☐ things = ☐ total

4 

☐ bags × ☐ things = ☐ total

5 

☐ bags × ☐ things = ☐ total

6 

☐ bags × ☐ things = ☐ total

**Code table**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 4 = J | 6 = U | 8 = R | 9 = M | 10 = E | 12 = B | 15 = P | 16 = N |

**Check it.** Write the letters in order.
Do they spell a word? ☐☐☐☐☐☐

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